

Transcriptional Reach of the Serine–Sphingomyelin–Membrane-Topology Axis in NK-Cell Immune Evasion: A Single-Cell Heterogeneous Graph Framework from Liver to Gastric Cancer

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Abstract

Motivation Zheng et al. (Nat Immunol 2023) showed that tumors evade natural killer (NK) cell cytotoxicity by depleting serine-derived sphingomyelin (SM) in NK membranes, collapsing membrane protrusions and abolishing lytic synapse formation – a mechanism proven with single-cell mass spectrometry and super-resolution imaging in a small liver-cancer cohort, methods that cannot scale to cohort-level analysis or be tested in other cancers.

Results We present GC-NKGraph-Atlas, which asks how much of this serine→sphingomyelin→membrane-topology→cytotoxicity axis is transcriptionally recoverable from public tumor transcriptomes. A mechanism-card formalism encodes the wet-lab mechanism as a machine-readable recipe driving cell-type-attributed proxy scoring, a mechanism-grounded heterogeneous gene graph, and a graph neural network (GNN) that predicts NK immune states. Across a two-arm design (liver positive control, gastric extension) we obtain a three-layer scoping result. First, the effector arm (protrusion-machinery↔cytotoxicity) is detectable in bulk transcriptomes (TCGA-LIHC $r=0.55$; gastric replications GSE62254 $r=0.42$, GSE84437 $r=0.62$), but roughly half of this coupling reflects NK-cell-abundance covariation – after adjusting for an NK-fraction proxy it survives in two of three bulk cohorts, not the third, and the single-cell version of the same coupling does not survive residualization against library size/latent structure or a random-gene-module permutation baseline. We therefore anchor the effector-arm claim on the bulk result, treated as partially recoverable and substantially infiltration-driven. Second, transcription does not proxy the physical topology phenotype: intratumoral NK cells show higher, not lower, protrusion-machinery transcripts ($\Delta=+0.14$) despite reduced cytotoxic output ($\Delta=-0.14$) – opposite to the physical membrane collapse. Third, the upstream metabolic arm (SM-balance→protrusion) is not transcriptionally detectable at either resolution (bulk $r=-0.02$; single-cell corrected $r=0.03$, $P=0.20$), consistent with metabolite-level regulation. A second, independently authored mechanism card ($TGF\beta \rightarrow SMAD \rightarrow NK$ exclusion) reproduces the same boundary end-to-end, generalizing the finding across two distinct evasion mechanisms. The framework prioritizes 37 putative tumor-intrinsic candidate targets (led by PHGDH, SGMS2, PSAT1, PSPH, SMPD3/1), each with a recommended validation assay, presented as a hypothesis-generation shortlist rather than a tumor-exclusive target set.

Availability All code, configuration, and synthetic test data are available at <https://github.com/nblvguohao/GC-NKGraph-Atlas>. The mechanism-card template enables application to additional published mechanisms without modifying the pipeline core.

Keywords NK cell; immune evasion; heterogeneous graph neural network; single-cell transcriptomics; sphingomyelin; gastric cancer

Key Points

- A **mechanism-card** formalism converts published wet-lab immune-evasion mechanisms into scalable transcriptome analyses; demonstrated end-to-end on two independent

cards (serine–SM; $TGF\beta \rightarrow SMAD$) that reveal a consistent transcriptional-reach boundary.

- A **two-arm design** (liver positive control + gastric extension) shows the effector arm (protrusion→cytotoxicity) is recoverable in bulk transcriptomes but **~50% of the coupling is NK-abundance covariation**, and the single-cell coupling does not survive technical-confound controls; the metabolic arm is not detectable after correction ($p=0.20$).
- **The reach boundary generalizes**: a second, independently authored card reproduces the same pattern – generic NK-effector co-variation recovers, but mechanism-specific causal predictions fail in bulk.
- **Transcription does not proxy the physical topology phenotype**: intratumoral NK cells show higher protrusion-machinery transcripts despite collapsed physical protrusions – a fundamental disconnect defining the boundary of transcriptome-based reconstruction.
- The graph is used as a **probe, not a predictor**: ablating the `metabolic_crosstalk` edge abolishes the corresponding embedding coupling, converging with the null correlation result; the framework outputs a de-circularized 37-gene candidate list with recommended validation assays.

1. Introduction

Natural killer (NK) cells kill tumor cells without prior antigen sensitization, making them attractive cancer-immunotherapy effectors [1–5], but tumors evade NK killing through metabolic competition, inhibitory-ligand upregulation, immunosuppressive signaling, and physical exclusion [6–11]. Zheng et al. [12] elucidated a mechanism in which tumors dysregulate serine metabolism [13,14], depleting NK-membrane sphingomyelin (SM) [15,16], collapsing membrane protrusions, and abolishing lytic synapse formation – rescuable by SM-catabolism inhibition, synergistic with Tim3 blockade, and proposed to extend beyond liver cancer [17]. However, this mechanism was proven using super-resolution imaging and single-cell mass spectrometry in a small cohort with no deposited transcriptomic data – techniques that do not scale to cohort-level testing.

Three gaps motivate this work: (i) **scalability** – testing the axis across hundreds of samples requires a transcriptome-based proxy; (ii) **transferability** – whether the axis operates in gastric cancer, not on the mechanism’s published extension list, is unknown; (iii) **reusability** – each mechanistic discovery currently requires a bespoke computational follow-up.

We introduce **GC-NKGraph-Atlas**, built around a reusable **mechanism-card** abstraction: a machine-readable YAML specification encoding one published mechanism’s molecular chain, gene modules, expected directions, physical ground-truth targets, and validation logic. The card driving this study (`zheng_nk_sm_topology`) drives (a) cell-type-attributed transcriptional proxies from scRNA-seq, (b) a tumor–NK heterogeneous graph with a `metabolic_crosstalk` edge grounded in the specific biology, (c) a graph neural network learning gene embeddings to predict NK immune states, and (d) multi-evidence candidate-target ranking. A two-arm design tests **Arm A** (liver/HCC positive control, the system where the axis was proven) and **Arm B** (gastric-cancer extension).

Our framework draws on immune-state/cell-type inference tools (CIBERSORTx [18], quanTIseq [19], Scissor [20]), cell–cell-communication and metabolic-flux inference (CellChat

[22], CellPhoneDB [23], NicheNet [24], scFEA [25]), trajectory/dysfunction modeling (CytoTRACE [26]; CREM regulation [27]), and graph learning in cancer genomics [28–32], including genome-scale driver-gene-discovery fusion models such as TREE [49] and GRAFT [50]. None of these operationalizes a single published mechanism as a machine-readable card with pre-registered claim boundaries between transcriptional proxy and physical phenotype, nor asks – rather than assumes – how far a specific mechanism’s transcriptional signal reaches; TREE and GRAFT in particular target genome-scale, cancer-agnostic driver discovery from generic networks, a different scale, label, and objective from this study’s ~100-gene, mechanism-anchored, reach-mapping design (§4).

2. Methods

Study design. The pipeline runs 14 phases across five stages: data acquisition (TCGA-STAD/LIHC, GEO gastric cohorts), scRNA integration and NK-state scoring, heterogeneous graph construction, GNN model training against six tabular baselines, and target prioritization (`src/pipeline.py`, checkpoint-based, with a synthetic-data test mode).

Mechanism-card abstraction. A mechanism card is a YAML specification (schema: `origin`, `biology`, `transcriptional_proxy`, `physical_ground_truth` [gated], `graph_integration`, `validation`, `therapeutic_hook`; full schema in `configs/mechanism_cards/mechanism_card.template.yaml`) that separates what is computable from transcriptomes from what requires physical measurement; mock data are excluded from all scientific claims.

SST-axis transcriptional proxy. The serine–SM–topology axis is operationalized as seven gene modules (~100 genes: tumor serine capacity, NK SM synthesis/catabolism, NK de-novo sphingolipid, NK protrusion machinery, NK synapse/cytotoxicity outcome, checkpoint link) yielding derived scores `nk_sm_balance`, `nk_topology_permissive`, and `sst_axis_score`. Claims are strictly limited to “transcriptional program permissive-of/associated-with” the topology phenotype; physical-topology or metabolite-content prediction is prohibited.

scRNA-seq analysis. GSE246662 (nine samples, healthy liver/gastric cancer/liver metastasis, 166,829 cells) was library-size normalized, scVI-integrated (30 latent dims), and Leiden-clustered (SCANPY [33–37]); NK cells ($n=8,310$) were separated from T/monocyte/B lineages by marker threshold. Four NK states (hot-cytotoxic, hot-dysfunctional, cold/excluded, intermediate) were defined from cytotoxicity/dysfunction/exclusion gene-set z-scores [39,40] and projected onto bulk samples by scRNA-anchored scoring. No hard per-cell QC threshold was applied (Limitations); technical variance is addressed via count-depth residualization (§3.2).

Heterogeneous graph. Nodes are the ~100 SST-axis/candidate genes; edges combine standard priors (STRING PPI, CellChatDB ligand–receptor) with two mechanism-grounded types (`metabolic_crosstalk`: tumor-serine gene to NK-topology gene, weight 0.5, fixed not fit to data; `sm_topology_axis`) and scRNA co-expression. The `metabolic_crosstalk` edge is diagnostic, not predictive: it lets us ask whether hard-coding the mechanism’s upstream coupling is corroborated by the data (§3.7).

Model. Gene embeddings are learned by truncated-SVD spectral decomposition of the combined adjacency matrix (all

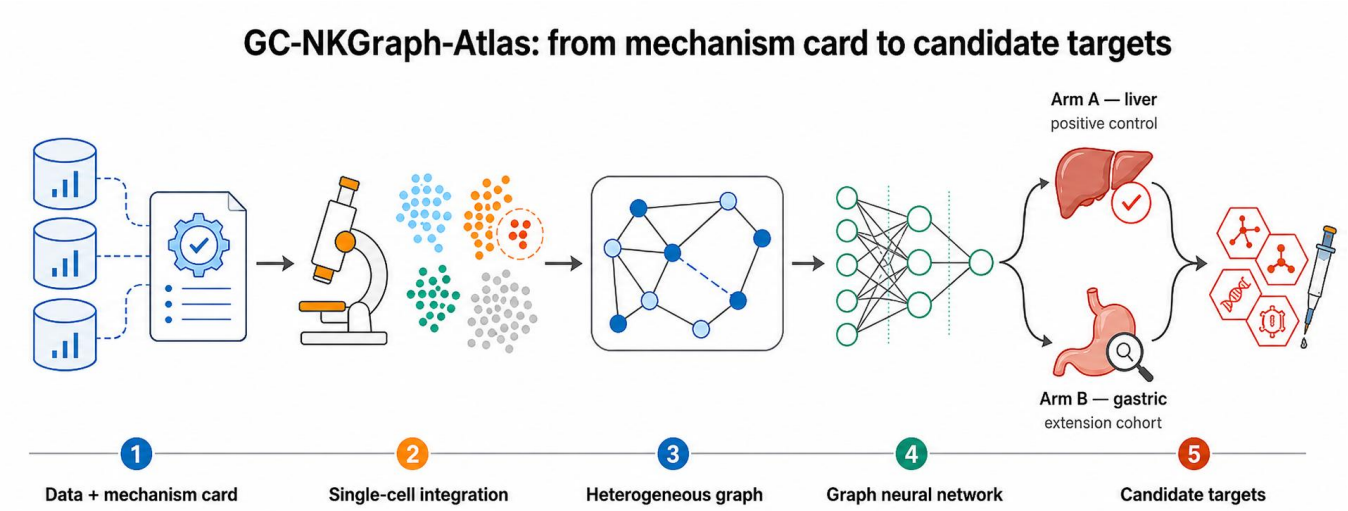


Figure 1 Workflow overview of the GC-NKGraph-Atlas framework, from data acquisition through target prioritization, showing the mechanism-card abstraction and the two-arm study design (Arm A: liver positive control; Arm B: gastric cancer extension). *Alt text: Workflow overview of the GC-NKGraph-Atlas framework showing the computational pipeline stages, the mechanism-card abstraction, and the two-arm study design.*

Table 1. Pre-registered hypotheses.

ID	Hypothesis	Expected direction
H1	tumor_serine_capacity ⊥ nk_sm_balance	Direction calibrated, reported
H2	nk_sm_balance (+) nk_protrusion_machinery	Positive
H3	nk_protrusion_machinery (+) cytotoxicity anchors	Positive
H4	nk_topology_permissive (−) HAVCR2 / dysfunction	Negative
H5	intratumoral NK i peritumoral/peripheral NK in sm_balance & protrusion machinery (scRNA)	Negative (tumor i normal)

reported results, including the ablation, use this encoder; a learnable heterogeneous graph transformer [32, 41] was not used – §4.4). A classifier MLP on [expression, expression-embedding] predicts NK state (5-fold stratified CV; Adam, hyperparameters selected by a 100-trial Bayesian search). Six tabular baselines (XGBoost [42], LightGBM [43], Random Forest, ElasticNet, SVM, MLP) and two domain-knowledge baselines (NK-marker signature; SST-module signature) are evaluated on identical splits. A six-view label-masked multi-view audit (generic/LR/metabolic_crosstalk/sm_topology_axis/GO/MSigDB views [49,50], spectral SVD, global-softmax fusion) tests whether learned view-fusion adds predictive value once every NK-state-label-defining gene is masked from inputs, using ten fixed seeds (STAD 80:20 train/early-stop, LIHC held out).

Target prioritization. Candidate genes are ranked by a five-dimension composite (tumor specificity 0.30, NK-dysfunction correlation 0.20, SST-axis membership 0.30, axis-core membership 0.10, literature support 0.10); tumor-intrinsic candidates are required to be up in malignant cells, separating them from NK-side axis-readout genes.

Pre-registered hypotheses and pseudoreplication correction.

Five hypotheses were pre-registered for Arm A with a full-recovery criterion requiring H2–H5 to pass. Single-cell NK correlations (8,310 cells, 9 samples) are corrected for

pseudoreplication by per-sample Pearson r , Fisher- z transform, and DerSimonian-Laird random-effects meta-analysis across samples; corrected values are reported throughout (naive per-cell values retained in results/tables/sst_axis_pseudoreplication_corrected.tsv for transparency). Full protocol detail (QC thresholds, scVI/Leiden parameters, hyperparameter search, and the multi-view audit design) is in Supplementary Methods S.M.1–S.M.5.

3. Results

3.1. Dataset summary

Four bulk cohorts and one multi-tissue scRNA dataset were processed (Table 2): TCGA-LIHC ($n=423$, liver, positive control), TCGA-STAD ($n=450$, gastric, primary training [44]), GSE62254/ACRG ($n=300$ [45]) and GSE84437 ($n=483$, gastric, external validation; both required probe-to-symbol remapping), and GSE246662 (166,829 cells, 8,310 NK). Bulk NK-state labels were assigned by scRNA-anchored scoring.

Table 2. Dataset characteristics.

Dataset	Cancer	Type	Samples / cells	Role
TCGA-LIHC	Liver	Bulk RNA-seq	423	Positive control (Arm A)
TCGA-STAD	Gastric	Bulk RNA-seq	450	Train primary (Arm B)
GSE62254 (ACRG)	Gastric	Bulk microarray	300	External validation
GSE84437	Gastric	Bulk microarray	483	External validation
GSE246662	Liver/gastric/metastasis	scRNA-seq	166,829 cells (8,310 NK)	NK atlas + axis test

3.2. Arm A – the axis is partially recovered: effector arm yes, topology arm no

Hypotheses H1–H5 were tested at bulk and single-cell resolution (Table 3; Figure 2). **H1** (serine-capacity↔SM-balance) is null at both resolutions (bulk $r=-0.02$, $P=0.74$; single-cell $r=+0.01$, $P=0.27$), consistent with serine→SM crosstalk operating at the metabolite rather than transcript level. **H3** (protrusion↔cytotoxicity, the effector arm) recovers in bulk TCGA-LIHC ($r=0.55$, $P=5\times10^{-35}$), but two controls qualify this. First, roughly half of the bulk coupling is NK-cell-abundance covariation: partialling out a clean NK-lineage-marker proxy attenuates it to a partial $r=0.25$ in TCGA-LIHC, holding in two of three bulk cohorts but abolished in the third (GSE84437, partial $r=-0.07$, n.s.). Second, the same coupling at single-cell resolution (corrected meta-analytic $r=0.31$ across 9 samples) does not survive further technical-confound controls: it collapses to $r=0.09$ after residualizing against library size and the real scVI latent space, and does not exceed a random-gene-module permutation null (empirical $P=0.97$). We therefore anchor the effector-arm claim on the bulk result and its gastric replications (§3.3), and report the effector arm as *partially recoverable and substantially infiltration-driven*. **H2** (SM-balance→protrusion, the metabolic arm) is undetectable in bulk ($r=-0.02$, $P=0.72$) and not significant after pseudoreplication correction at single-cell resolution (corrected $r=0.03$, $P=0.20$). **H4/H5** (physical topology) show the most consequential result: intratumoral NK cells have *higher*, not lower, protrusion-machinery transcripts than normal-tissue NK ($\Delta=+0.14$, $P=3\times10^{-91}$) despite correctly reduced cytotoxic output ($\Delta=-0.14$, $P=6\times10^{-52}$) – opposite to the physical membrane collapse, externally corroborated for the cytotoxicity direction [46]. This is the natural boundary of transcriptome-based reconstruction for a mechanism whose phenotype is physical and whose causal link operates at the metabolite level, precisely why the anchor study required single-cell mass spectrometry and super-resolution imaging. Full derivation for H2–H5 is in Supplementary Results S3.2.

3.3. Arm B – Gastric cancer extension

The effector coupling extends to gastric cancer: scRNA ($r=0.35$, $P=7\times10^{-30}$, $n=1,017$ gastric NK, sharing the same single-cell technical caveats as §3.2) and, after platform-annotation remapping (GSE62254→GPL570; GSE84437→GPL6947), both external microarray cohorts (Table 4). As in Arm A, this bulk replication is substantially NK-abundance-driven (§3.2): after partialling out NK fraction it remains significant in GSE62254 but not GSE84437. The metabolic coupling (H2), null in liver, is weakly but significantly positive in both gastric bulk cohorts

($r=0.18/0.11$), consistent with a real but small effect more detectable where NK signal is stronger. Gastric-cancer NK show the lowest cytotoxicity-output score of the three tissues sampled, consistent with an evasion phenotype (Figure 3).

3.4. NK immune state classification

On TCGA-STAD (5-fold CV, NK-hot-cytotoxic vs rest), the graph-informed GNN (MCC 0.706, AUROC 0.950) is statistically indistinguishable from the top gradient-boosting baselines (LightGBM MCC 0.733, XGBoost 0.727; paired $P_{\chi^2}0.28$) and from a no-graph SST-module-signature baseline (MCC 0.619, AUROC 0.904; $P_{\chi^2}0.10$), while significantly exceeding a simple NK-marker-signature baseline and the linear/kernel/shallow-network baselines (Table 5, Figure 4). The graph therefore does not provide a discriminative advantage over simpler alternatives; its value here is diagnostic (§3.7), not predictive. Full baseline detail, including a CIBERSORTx/quantiseq/Scissor comparison roadmap, is in Supplementary Results S3.4.

3.5. Candidate target prioritization (de-circularized)

Separating tumor-intrinsic candidates (required up in malignant cells) from NK-side axis-readout genes yields 37 tumor-intrinsic candidates, headed by the druggable serine/SM enzymes PHGDH, SGMS2, PSAT1, PSPH, and SMPD3/1 (Table 6). A trivial mechanism-card-membership-only baseline correlates moderately with the five-dimension scoring (Spearman $\rho=0.54$): the added value is re-ordering by tumor specificity and surfacing 12 gastric-relevant non-SST genes (e.g., COL1A1/2, NECTIN2, ERBB2, FGFR2, MET). Two orthogonal filters – NK-state differential expression (externally replicated in two gastric cohorts) and DepMap 26Q1 CRISPR essentiality in 35 gastric cell lines – refine the list into evidence tiers (Table 6): **SMPD3/SMPD1** pass both filters cleanly; **SGMS2** and **NT5E/CD73** pass on distinct evidence but carry caveats (discordant external DE; label self-referentiality, respectively); five serine-pathway genes and RAC1/ERBB2 are downgraded or excluded for wrong-direction DE or pan-essentiality. The list is de-circularized but not fully clean (46% of candidates retain NK-side module annotation) and is a hypothesis-generation shortlist, not a validated tumor-exclusive set. Full evidence and external replication are in Supplementary Results S3.5.

3.7. The graph as a convergent probe of transcriptional reach

Ablating the metabolic_crosstalk edge from the enriched heterogeneous graph abolishes the corresponding embedding coupling (H1 embedding-coupling $0.128\rightarrow -0.002$; Table 7)

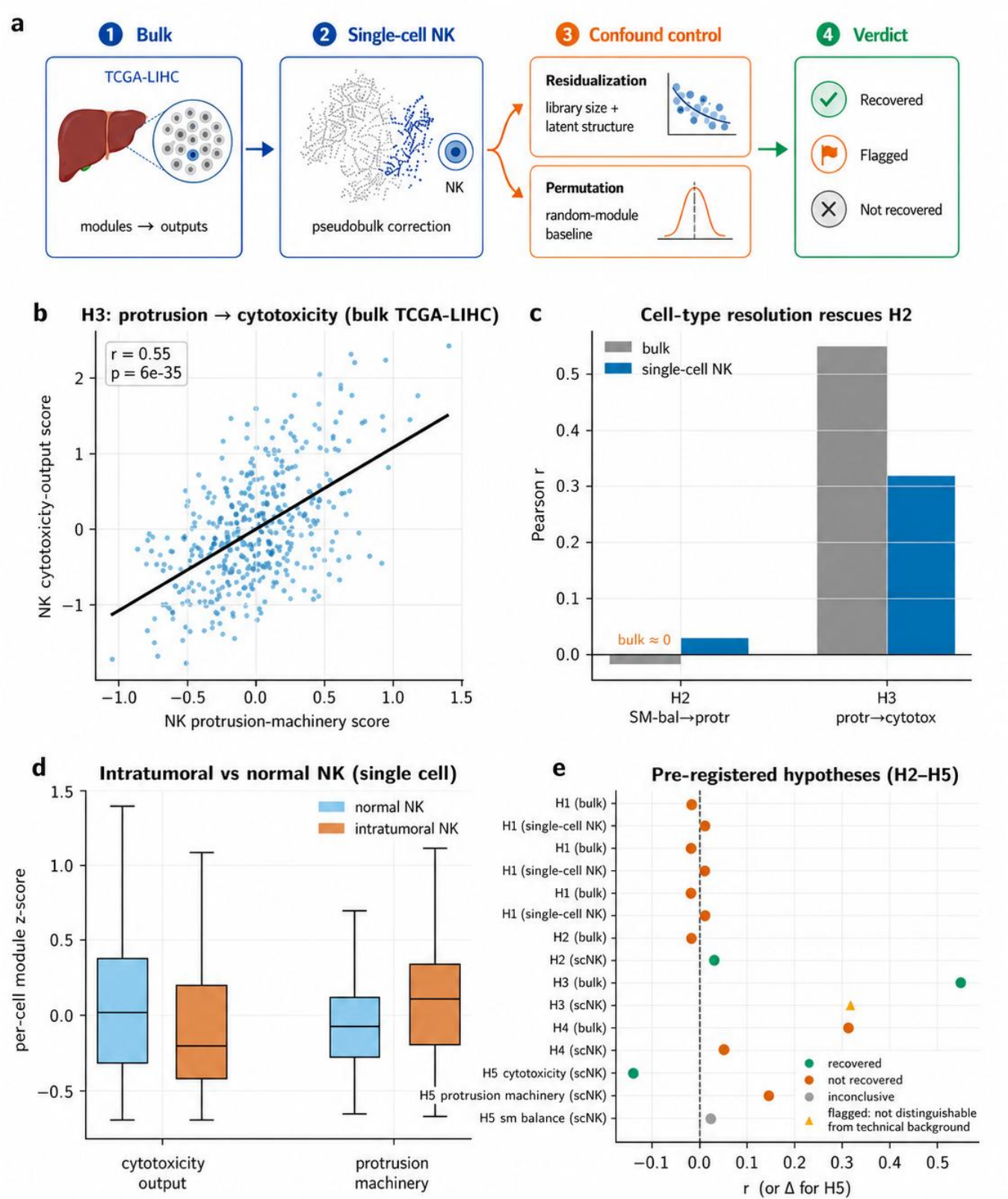


Figure 2 (a) Confound-control logic for the effector-arm claim. (b) H3: protrusion-machinery vs cytotoxicity-output score, bulk TCGA-LIHC ($r=0.55$, $p=5\times 10^{-35}$). (c) H2 and H3 at bulk vs single-cell resolution. (d) Intratumoral vs normal NK per-cell module z-scores. (e) Table 3 as a forest plot. *Alt text: Five-panel summary of liver positive-control recovery.*

Table 3. Pre-registered hypothesis outcomes (multi-resolution). † Single-cell NK r/P are meta-analytic values after pseudoreplication correction (Methods §2.9). ‡ The H3 single-cell coupling additionally fails count-depth/scVI-latent residualization and a random-module permutation test (empirical $P=0.97$); full battery in Supplementary Results S3.2. The effector-arm claim rests on the bulk result (this table) and its bulk gastric replications (§3.3), not the single-cell H3 number.

Hyp	Test	Resolution	r / Δ	P	Expected	Outcome
H1	serine_cap sm_balance	~ bulk	−0.016	0.74	calibrated	reported (null)
H1	serine_cap sm_balance	~ single-cell NK	+0.012	0.27	calibrated	reported (null)
H2	sm_balance ~ protrusion	bulk	−0.017	0.72	+	not recovered
H2	sm_balance ~ protrusion	single-cell NK	+0.029	0.20	+	not signif. after correction
H3	protrusion ~ cytotoxicity	bulk	+0.551	5×10^{-35}	+	recovered
H3	protrusion ~ cytotoxicity	single-cell NK	+0.313†‡	3.9×10^{-8} †‡	+	not distinguishable from technical background
H4	topology ~ dysfunction	bulk	+0.311	7×10^{-11}	−	not recovered
H4	topology ~ HAVCR2	single-cell NK	+0.036†	9.1×10^{-4} †	−	not recovered (after corr.)
H5	intratumoral normal: cytotoxicity	single-cell NK	−0.141	6×10^{-52}	tumor	recovered
H5	intratumoral normal: protrusion	single-cell NK	+0.142	3×10^{-91}	tumor	not recovered

Table 4. External validation of the axis (independent gastric cohorts).

Cohort	n	protrusion~cytotoxicity r (p)	sm_balance~protrusion r (p)	NK markers
GSE62254 (GPL570)	300	0.42 (1.4×10^{-14})	0.18 (1.3×10^{-3})	6/7
GSE84437 (GPL6947)	483	0.62 (3.3×10^{-53})	0.11 (0.02)	7/7

while leaving the within-axis `sm_topology_axis` coupling comparatively intact – an independent, architecture-based confirmation that the upstream metabolic coupling is transcriptionally absent, converging with the null H1/H2 correlations (§3.2). In the label-masked multi-view audit, learned view-fusion is significantly *worse* than a no-graph baseline for both AUROC and AUPRC ($\Delta = -0.010 / -0.021$, bootstrap CI excluding 0), and no view is stably top-ranked across seeds – the unmasked sensitivity analysis reaches much higher accuracy (0.971/0.936), confirming that label overlap, not graph structure, drove the earlier optimistic result. A node-label permutation test confirms both mechanism-grounded edge types impose their intended geometry (both empirical $P = 0.001$), verifying structural specificity without implying biological or predictive validation. Together, three methodologically independent lines of evidence – correlation, graph ablation, and multi-view audit – converge on the same boundary: the effector layer is transcriptionally present; the upstream metabolic coupling is not. Full multi-view diagnostics are in Supplementary Results S3.7 and Fig. S1.

Beyond the two mechanism cards run end-to-end above, an exploratory scan applying the same comparisons to all

four registered cards (including the two validated only structurally) across additional cohorts and modalities is reported in Supplementary Results S3.8 and Fig. S2; its pre-registered cross-mechanism gate is not met (`comparative_atlas_only`), so it is not treated as further evidence for or against the reach boundary above – only the two-card result in §3.7/§4.1 is.

4. Discussion

4.1. Summary of findings

Our central result is a **map of the mechanism’s transcriptional reach**, not a blanket claim of recovery. Four findings anchor it:

1. **The effector layer is transcriptionally recoverable in bulk transcriptomes and generalizes to gastric cancer**, but is substantially NK-abundance-driven: the protrusion-machinery→cytotoxicity coupling replicates from independent liver ($r=0.55$) and gastric transcriptomes, yet after adjusting for NK-fraction it attenuates by ~50%, holding in two of three bulk cohorts and not the third. The single-cell

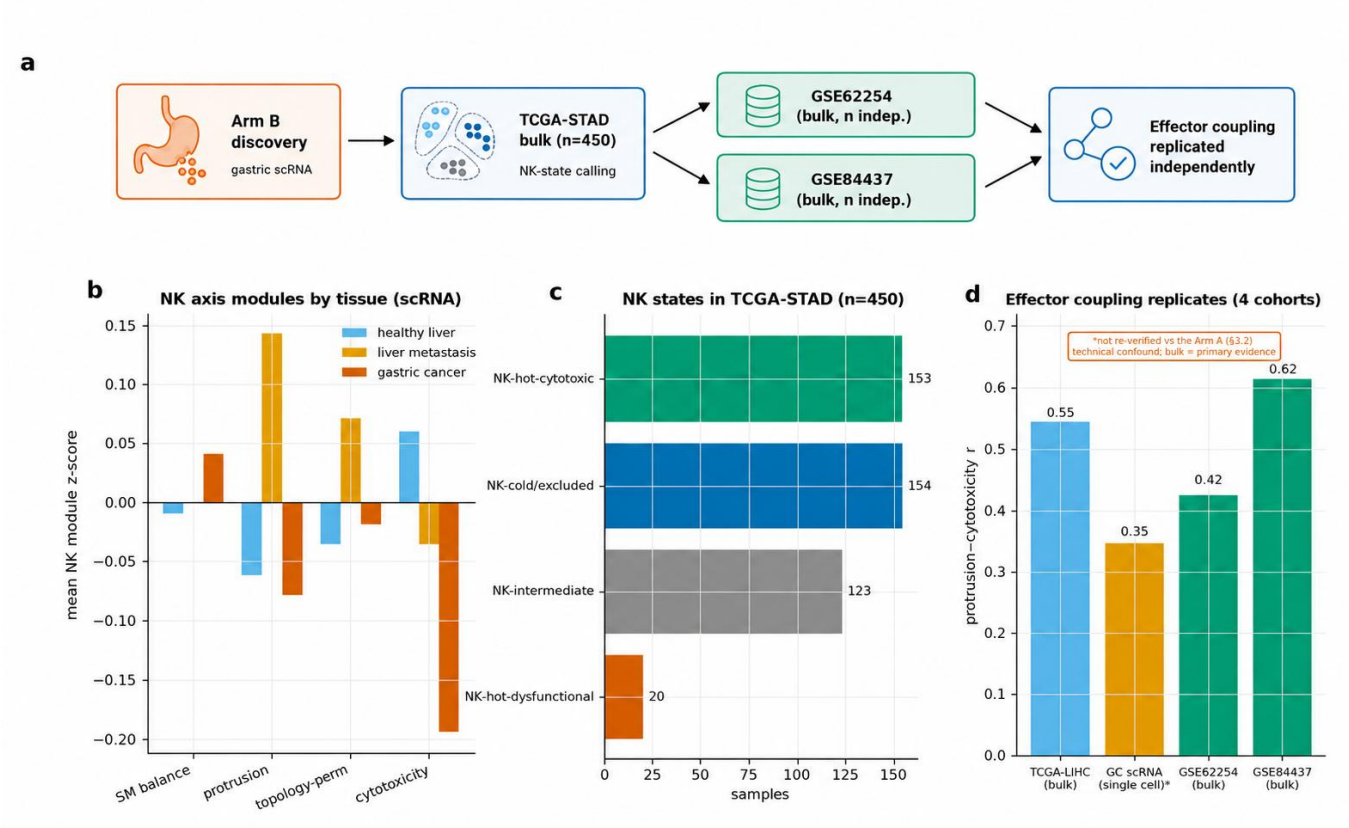


Figure 3 (a) External-validation design. (b) Cross-tissue module comparison. (c) NK-immune-state distribution in TCGA-STAD (n=450). (d) Effector-coupling replication across cohorts. *Alt text: Four-panel gastric-cancer extension figure.*

Table 5. NK-state classification (TCGA-STAD, 5-fold CV; mean over folds). GNN not significantly different from LightGBM/XGBoost (paired $p>0.27$) or the SST-module signature baseline ($p>0.10$); significantly above the NK-marker signature baseline and ElasticNet/SVM/MLP ($p<0.05$).

Method	Accuracy	Balanced Acc.	Macro F1	MCC	AUROC	AUPRC
LightGBM	0.880	0.863	0.865	0.733	0.960	0.933
XGBoost	0.878	0.859	0.862	0.727	0.959	0.928
GC- NKGraph- Atlas (GNN)	0.864	0.856	0.850	0.706	0.950	0.910
RandomForest	0.860	0.825	0.836	0.681	0.941	0.901
ElasticNet	0.856	0.822	0.832	0.671	0.927	0.884
SST- module signature	0.818	0.823	0.803	0.619	0.904	0.827
NK- marker signature	0.787	0.744	0.750	0.503	0.849	0.733
SVM (RBF)	0.773	0.672	0.683	0.489	0.900	0.839
MLP (2- layer)	0.747	0.669	0.641	0.376	0.807	0.730

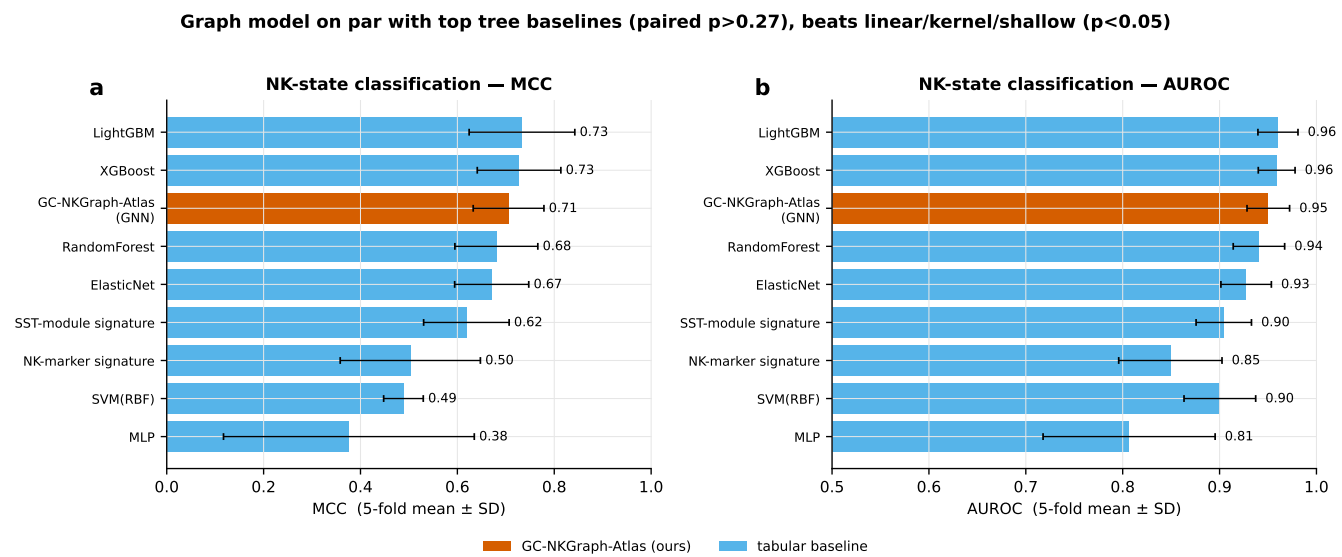


Figure 4 MCC (a) and AUROC (b) for GC-NKGraph-Atlas versus tabular and signature baselines on identical seed-42 folds; error bars show cross-fold SD. *Alt text: Bar charts of MCC and AUROC comparing the graph model with tabular and signature baselines.*

Table 6. Top putative tumor-intrinsic candidate targets in gastric cancer, evidence-tiered. Tier 1: non-pan-essential plus NK-state DE or mechanistic privilege. Tier 2: single-dimension support. Tier 3: downgraded (NK-state signal absent/opposite). Tier X: excluded (pan/moderately essential). § SGMS2's DE signal is discordant in external replication. ‡ NT5E's DE result is partly self-referential (Supplementary Results S3.5).

Tier	Gene	Category	tumor_spec (log2FC)	NK-state DE	DepMap CERES (real, 35 gastric lines)	Druggability
1§	SGMS2	SM synthesis	+0.038	UP in dysf (FDR=0.007); discordant externally	weakly essential (−0.02)	Preclinical
1‡	NT5E	adenosine (CD73)	+0.013	UP in dysf (FDR=0.014)	non-essential (+0.005)	Phase 1/2 clinical
1	SMPD3	SM catabolism	+0.032	no signal	weakly essential (−0.13)	Preclinical (Zheng 2023)
1	SMPD1	SM catabolism	+0.001	no signal	weakly essential (−0.16)	Preclinical (Zheng 2023)
2	PHGDH	serine metabolism	+0.059	no signal	weakly essential (−0.04)	Phase 1/2 (PHGDH inhibitor)
2	MICA	stress ligand	+0.006	no signal	non-essential (+0.05)	Preclinical
3	PSAT1	serine metabolism	+0.016	no signal	moderately essential (−0.27)	– downgraded
3	PSPH	serine metabolism	+0.010	DOWN in dysf (p=0.03)	weakly essential (−0.16)	– downgraded
3	SHMT1/2, MTHFD1L	serine/1C metabolism	+0.001–0.008	DOWN in dysf (all p<0.02)	weakly essential (−0.10 each)	– downgraded
X	MTHFD1	serine/1C metabolism	+0.008	DOWN in dysf (p<0.02)	pan-essential (−0.53)	– not a target
X	ERBB2	gastric oncogene	+0.069	DOWN trend	moderately essential (−0.36)	FDA approved (HER2+ GC), but a confound
X	RAC1	NK protrusion GTPase	+0.077	DOWN in dysf (p=0.04)	pan-essential (−0.74)	– not a target

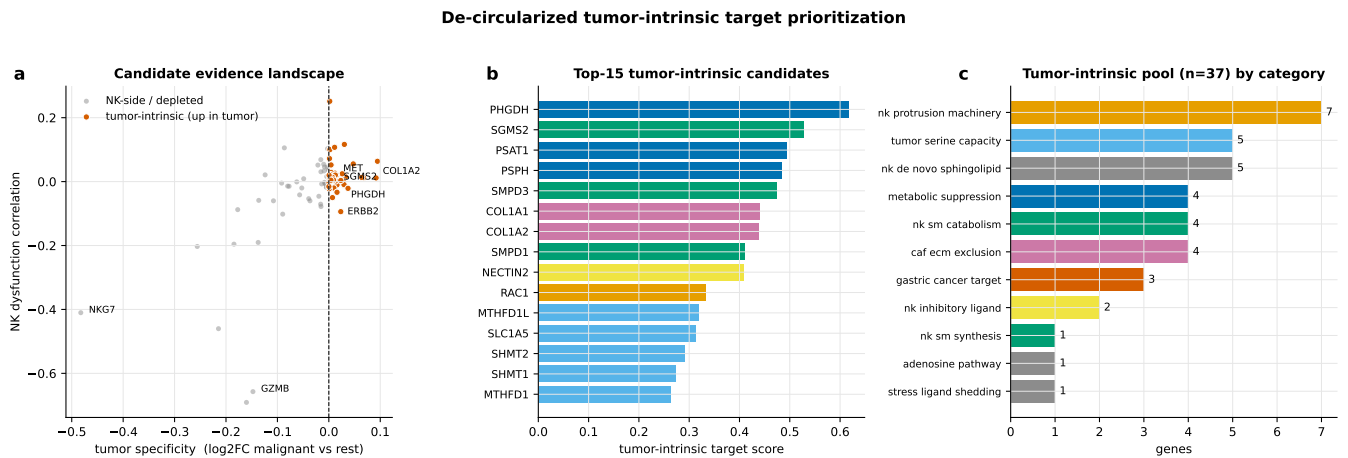


Figure 5 (a) Tumor-specificity vs NK-dysfunction-correlation scatter. (b) Top-15 tumor-intrinsic candidates. (c) Category composition of the 37-gene pool. *Alt text: Three-panel target-prioritization figure.*

Table 7. Graph ablation on the enriched real graph. Embedding-coupling = mean cosine similarity between module gene embeddings; modularity = intra-module minus inter-module cosine similarity.

Variant	Edges	Modularity	H1 embedding-coupling (serine↔SM)	H2 embedding-coupling (SM↔protrusion)
FULL (all edges)	1,612	0.279	0.128	0.183
–MC (without metabolic_crosstalk)	1,312	0.288	–0.002	0.146
–SST (mechanism edges removed; generic priors only)	492	0.287	–0.002	–0.000

version of the same coupling does not survive residualization against library size/latent structure or a random-module permutation baseline (§3.2), so we anchor this finding on the bulk result.

- The metabolic coupling is not transcriptionally detectable – by two independent methods.** Correlation analysis (H1, H2 both null after correction) and graph ablation (removing the `metabolic_crosstalk` edge abolishes the imposed embedding coupling, §3.7) converge: this coupling is real biology operating at the metabolite level, not the transcript level, exactly the boundary the mechanism card’s “gated physical ground-truth” section is designed to mark.
- Transcription does not substitute for the physical topology phenotype – a finding, not a failure.** Machinery-gene transcription runs opposite to the physical protrusion collapse in intratumoral NK (higher transcript, lower function), reproducing why the anchor lab required single-cell mass spectrometry and super-resolution imaging, and identifying the post-transcriptional membrane-recruitment layer as the critical gap for future experimental work.
- The transcriptional-reach boundary generalizes across mechanisms.** A second, independently authored mechanism card ($\text{TGF}\beta \rightarrow \text{SMAD} \rightarrow \text{NK exclusion}$ [10]) run end-to-end on the same cohorts reproduces the pattern: NK-effector/outcome co-variation is recoverable, but the mechanism-specific causal predictions ($\text{TGF}\beta$ signaling suppresses activating receptors; CAF-ECM excludes cytotoxic NK) fail in bulk, once NK-cell fraction is controlled – the same shared cause as the Zheng card’s upstream metabolic coupling failing.

Independent orthogonal validation (NK-state DE + DepMap essentiality, §3.5) refines the 37-gene candidate list into evidence tiers: SMPD3 and SMPD1 pass both filters cleanly; SGMS2 and NT5E/CD73 pass with caveats (discordant external DE and label self-referentiality, respectively); five serine-pathway genes and ERBB2/RAC1 are downgraded or excluded. A per-sample readout combining NK SM-catabolism score and HAVCR2 (Tim3) expression can stratify samples by the SM-restoration + Tim3-blockade combination logic proposed in the anchor paper – an in-silico hypothesis for experimental testing, not a validated clinical predictor; this is consistent with an independent HCC study reporting co-regulation of NK checkpoints (HAVCR2, TIGIT, PDCD1, LAG-3, CLEC12B) [46].

4.2. The mechanism-card approach

A key contribution is the mechanism-card abstraction itself: a machine-readable YAML formalism separating a mechanism’s computational operationalization from the pipeline that executes it, so applying the framework to a new mechanism requires only authoring a new card, not rewriting the pipeline core. Beyond reuse, the card format enforces scientific discipline shown here to be load-bearing rather than decorative: explicit claim boundaries (letting us state without ambiguity that machinery transcription does not proxy physical topology), gated physical ground-truth targets (preventing conflation of transcriptional proxy with direct measurement), and pre-registered hypotheses with a defined recovery criterion (so Arm A’s partial outcome is reported as a structured result rather than reframed).

The mechanism-card registry holds four cards (serine-SM, adenosine-mediated suppression, $\text{TGF}\beta$ -driven exclusion,

MICA/B shedding) sharing one YAML schema and low pairwise gene overlap (Jaccard 0.07–0.19). **Two have now been run end-to-end on real data:** the $TGF\beta \rightarrow SMAD \rightarrow NK$ -exclusion card reproduces the same transcriptional-reach boundary as the Zheng card rather than a card-specific artifact – effector/outcome co-variation recovers, but its mechanism-specific causal predictions fail in bulk. The adenosine and MICA/B cards remain structurally validated but not yet run end-to-end. Reusability is therefore an established capability, not only a design aspiration.

4.3. Limitations

The complete 17-item limitations discussion is in Supplementary Results S4.3; key points are summarized here.

1. **Transcriptional proxy $\neq$ physical topology**, and serine/SM crosstalk is a metabolite-level effect not captured by transcription – both now empirically supported by the H1/H2 null results.
2. **No experimental validation**; all targets are computationally prioritized, with recommended assays bridging to a companion wet-lab program.
3. **Single-cell pseudoreplication and technical confounding**. The H3 single-cell coupling requires count-depth/module-permutation controls beyond pseudoreplication correction alone (§3.2), and does not survive them; a leave-one-sample-out check shows the pooled estimate itself is not an artifact of one sample.
4. **The graph does not outperform top tabular or signature baselines on accuracy** – confirmed by two independent stricter audits (§3.4, §3.7); an exploratory single-edge cross-cohort transfer test proved highly seed-sensitive and supports no directional claim.
5. **Residual NK-side bias in the tumor-intrinsic candidate pool** (46% retain NK-side module annotation) and **omission of intracellular/TF-level regulators** (e.g. CREM [27]).
6. **No clinical-outcome anchor**; NK states are linked to scRNA-defined labels, not survival or therapy response.
7. **Multi-card validation covers two of four cards end-to-end**; the bulk effector coupling is substantially NK-abundance-driven (§3.2); DepMap essentiality uses the real, current 26Q1 release; NK-state DE is underpowered for the dysfunctional group ($n=20$) though 88% externally concordant.
8. **The NK-state classification label partly overlaps the classifier's own input features** (a gene-set separation audit found label-defining marker genes are also present in the input expression vector), so §3.4's classification accuracy should be read accordingly; the SST-axis mechanism tests (H1–H5) do not share this issue.
9. **No hard per-cell QC threshold was applied** to the scRNA-seq data; technical variance is instead addressed via count-depth residualization.

4.4. Future directions

- **Additional cancer types and mechanisms**. Other digestive-tract cancers (colorectal, pancreatic, esophageal)

using the same card; the registry already carries authored-but-not-run cards for adenosine suppression, MICA/B-ADAM17 shedding, two checkpoint axes [47,48], and two metabolite axes, plus a not-yet-authored CLEC12B-lipoprotein-lipase axis [46].

- **Physical topology integration** once membrane protrusion/microvilli imaging data become available, and a **prospective validation cohort** with paired bulk RNA-seq, scRNA-seq, and functional NK assays.
- **Genome-scale learnable encoder** (per-node GCN/Transformer as in TREE [49] or GRAFT [50]), reserved for a genome-scale panel and an independently powered endpoint free of the small-panel overfitting risk exposed here.

5. Conclusion

GC-NKGraph-Atlas demonstrates *how far* a specific, published immune-evasion mechanism – the serine-sphingomyelin-membrane-topology axis of NK dysfunction – can be surveyed at cohort scale from public transcriptomes. Using a single-cell-informed heterogeneous graph framework and a two-arm design, we establish a three-layer scoping map: (i) the effector arm is recoverable in bulk transcriptomes from liver to gastric cancer, but ~50% of the bulk coupling is NK-cell-abundance covariation and the single-cell version does not survive technical-confound controls, so we report it as partially recoverable and substantially infiltration-driven rather than robustly recovered; this boundary reproduces on a second, independently authored mechanism card ($TGF\beta \rightarrow SMAD \rightarrow NK$ exclusion); (ii) the upstream metabolic coupling is **not** transcriptionally detectable after correction ($P=0.20$), consistent with metabolite-level regulation – independently confirmed by graph ablation; and (iii) the physical topology phenotype is fundamentally disconnected from machinery transcription – a finding, not a failure, that benchmarks the natural limit of transcriptome-based reconstruction for membrane-lipid mechanisms. This map, not an overclaim of full reconstruction, is the contribution.

The mechanism-card abstraction, demonstrated here on the Zheng 2023 serine-SM axis, provides a disciplined formalism for converting a published mechanism into a scalable study with explicit claim boundaries and gated physical ground-truth. This discipline is load-bearing: it is precisely the card's separation of computable-from-physical that lets us report the topology disconnection as a structured result, and the pre-registered recovery definition that forces the H2 non-significance to be reported plainly. The framework has now been run end-to-end on two independently authored cards, which converge on the same transcriptional-reach boundary; extending it to the remaining registered cards (adenosine-mediated NK suppression, MICA/B shedding) is a natural next step, so multi-card reuse is an established capability to be broadened rather than only a design aspiration.

Data Availability

All code and configuration files are available at <https://github.com/nblvguohao/GC-NKGraph-Atlas>. The mechanism-card template and the Zheng 2023 NK SM-topology card are provided under `configs/mechanism_cards/`. Synthetic test

data can be generated via `python src/common/synthetic_data.py`.

Reproducibility. The pipeline supports a fully self-contained synthetic data mode (≈ 17 MB, no real patient data) that exercises all phases end-to-end:

```
pip install -r requirements.txt
python src/common/synthetic_data.py # -> data/synthetic_data/
python src/pipeline.py --synthetic --force
```

An environment lock file (`environment.yml`) pins all dependency versions (testable on Linux/macOS/Windows with conda). Unit tests covering graph construction, NK scoring, SST-axis computation, and target prioritization are provided under `tests/` and can be run with `pytest tests/`. The pseudoreplication correction analysis is implemented in `src/topology/pseudoreplication_correction.py` and the count-depth control in `src/topology/count_depth_control.py`.

Real data availability:

- TCGA-STAD and TCGA-LIHC: available from the Genomic Data Commons (<https://portal.gdc.cancer.gov/>)
- GSE62254, GSE84437, GSE246662: available from the Gene Expression Omnibus (<https://www.ncbi.nlm.nih.gov/geo/>)
- DepMap Public 26Q1 (CRISPRGeneEffect.csv, Model.csv): the real, current release used for Table 6's CERES values, obtained directly from the DepMap portal at <https://depmap.org/portal/> (interactive download required – see Limitations). An earlier pass of this analysis used DepMap Public 24Q2 via the public figshare API (https://plus.figshare.com/articles/dataset/DepMap_24Q2_Public/25880521) before 26Q1 access was available; that snapshot has been superseded throughout.
- No novel sequencing data were generated for this study.

Pre-registration: The hypotheses H1–H5 and the original full-recovery criterion for the positive-control arm were registered in `configs/sst_axis_config.yaml` before execution. Because the full criterion was not met, the manuscript reports the result as a partial-recovery scoping map and explicitly separates pre-registered outcomes from post-hoc interpretation.

Author Contributions

Guohao Lyu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Yingchun Xia:** Methodology, Software, Validation, Writing – review & editing. **Huichao Liu:** Software, Data curation, Validation. **Xiaolei Zhu:** Formal analysis, Methodology, Writing – review & editing. **Shuai Yang:** Investigation, Validation, Visualization. **Ailian Zhou:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing. **Lichuan Gu:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing. All authors read and approved the final manuscript.

Ethics Approval and Consent to Participate

Not applicable. This study used only publicly available, de-identified datasets (TCGA-LIHC, TCGA-STAD, GSE62254, GSE84437, GSE246662) obtained from the Genomic Data Commons and the Gene Expression Omnibus. No new human or animal subjects were involved, and no identifiable personal data were generated or analyzed. Use of these public data complies with the respective data-access policies.

Competing Interests

The authors declare that they have no competing interests.

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